

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITTNA3-22

Group: 1

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Indicate	Yes	No
Plagiarism report attached		

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Section A

Question 1

1.1) The pandemic deepened inequalities that have an impact on innovation and research in African institutions, which includes:

- The digital divide between students when educational institutions provided e-learning to accommodate students during the pandemic. The digital divide in this context refers to certain students that live in areas with poor or no internet infrastructure where they weren't able to or had difficulties with e-learning, unlike the students who did have good internet infrastructure. Additionally, the 2020 General Household Survey (GHS) report of Stats SA indicated that only 63% of South Africans have access to the internet (Woldegiorgis, 2022). Therefore, any research being performed by the students with poor internet infrastructure had to be stopped until after the pandemic had ended and would need to prolong their studies in order to catch up on the learning that they missed out on during the pandemic's duration, which can prolong potential innovations from these students.
- The inability to make money during the pandemic for people who were laid off at their jobs. This can result in certain demographics being unable to continue paying for their studies at educational institutions. Moreover, people from areas like Ghana experienced layoffs and faced employment issues like underemployment during the pandemic where the money they make might not be enough to continue their studies in comparison to other students who have more money than them (Anas and Musah, 2025). Therefore, this can greatly hinder the full allocation of talent in an area due to an incomplete education, which can prevent potential innovations and research from occurring (Phiri *et al.*, 2016).
- The digital divide between students who utilise the educational institution's computing resources and the students that utilise their own computing resources. Students that rely on the resources provided by the educational institution were not able to continue their research and studies during the pandemic. Moreover, if the educational institution struggled during the pandemic, they could have lessened the amount of resources that they provide to students, so these students will have a more difficult time continuing with their studies or innovate in comparison to students with their own computational resources.
- The difference in skill sets between students from privileged and unprivileged homes that is produced by unequal e-learning. Therefore, the type of skill set that the student learns might not be compatible or lack requisite skills required for future labour market engagements, innovation and research (Anas and Musah, 2025).

However, these issues can be overcome by:

- To overcome the issues associated with the digital divide such as poor internet infrastructure or lack of access to computing devices can be reduced by the African union that can create a roadmap to implement easily accessible technology and technological infrastructure to various African countries (Audu and Gbadamosi, 2021).
- To assist with students that lack requisite skills, governments from various African countries can encourage individuals to participate in technological trainings that are state sponsored,

so that low-income students are able to participate in the training (Audu and Gbadamosi, 2021).

- Universities, if possible, can provide students who lack their own computing resources by providing devices like a laptop and provide internet connectivity in their residence if they lack it to enable them to utilise online learning tools. However, if universities are unable to provide assistance to their students, then the students can utilise NSFAS to receive funds that can enable them to continue their studies despite any financial hardships that they are experiencing, such as being laid off from their occupation (Woldegiorgis, 2022).

1.2) An emerging technology in Africa is Artificial Intelligence (AI) due to its ability to streamline the analysis of mass amounts of data. In addition, AI systems are being integrated into various industries, to improve upon existing technology or fill new roles, such as hospitals utilising AI to analyse medical images, echocardiography, or farms utilising predictive AI that provides suggested inputs and insurances policies based on the climate that the farm is experiencing (Badaru and Mphahlele, 2023).

Therefore, understanding how AI systems function and when to utilise them is an important skill that should be learned, because this emerging technology can improve the efficiency of existing systems as well as create new occupations centred around the developing and maintaining of AI systems, leading to economic growth. However, there are various issues that make up the large socio-economic issue that is the reluctant adoption of AI systems across Africa, which can be contributed to:

- The digital divide between urban and rural areas where people from rural areas with insufficient Information and Communication Technology (ICT) infrastructure, limited broadband access and unreliable power supplies are unable to fully utilise AI systems for their occupations, in comparison to people from urban areas.
- The inability to participate in training programmes related to AI systems in rural areas, where they are unable to improve their skill set that can be required of them from certain occupations.
- The lack of trust in AI systems with regards to data privacy and security due to the possibility of data breaches and the misuse of personal information (Maleka and Maidi, 2024).

Therefore, a project that can resolve this socio-economic issue should be conducted in order to for this emerging technology to be adopted effectively across Africa to improve potential economic growth. A project proposed to resolve this issue is the establishment of online infrastructure and training programmes in already existing, rural libraries to lower the digital divide between rural and urban areas. Local community members like teachers rely on the assistance provided by libraries for educational purposes, which demonstrates the importance of libraries for local communities. Therefore, the infrastructure of these libraries should be improved to provide centralised internet connectivity and computing resources where local community members are able to improve their overall digital literacy before participating in AI-related training programmes to enhance their skill set. Moreover, this project can collaborate with Non-Governmental Organisations (NGOs), universities and technology companies that can assist with funding and providing additional resources to these local communities, which could act as a catalyst for sustainable digital transformations (Israel Creleanor Mulaudzi and Ernest Kwesi Klu, 2026). In addition, the access to AI-related training programmes and computational resources can enable local community members to understand how it functions and how it should be utilised to mitigate security risks. This can improve trust in the technology among the community and potentially motivate them to incorporate AI technologies to assist them with education and work.

Moreover, a mentorship plan should be created that outlines strategies for sustainability and growth that the next generation of leaders can follow. An appropriate mentorship plan should consist of the following strategies:

- Current leaders should establish a culture that values continuous learning where future leaders will motivate themselves and other leaders to learn about new emerging technologies or socio-economic issues that can be resolved, which enhances sustainability and potential growth (Yani, Suherlan and Az Zaakiyyah, 2024).

- Mentors should utilise their experience and advise future leaders on important aspects such as the type of challenges that they are likely to face and how they should respond to it as well as how to identify and strategically plan for opportunities and trends that can result in growth, which can improve sustainability by solving issues quicker due to the advice (Yani, Suherlan and Az Zaakiyyah, 2024).
- Current leaders should enable future leaders to work with them on tasks in order for the future leaders to gain experience on how certain tasks should be performed, which improves sustainability and growth as these future leaders can utilise this experience when they mentor their next generation of leaders.
- Future leaders should perform their own independent research on aspects surrounding emerging technologies, like how it functions, any challenges it experiences and benefits that it provides, with guidance from mentors to ensure that the future leaders have the skills necessary to effectively identify, learn about and plan for emerging technologies. This can improve sustainability and growth by ensuring that any misconceptions or challenges that future leaders will face when they become leaders are addressed before they create issues within the programme.

1.3) Japan is facing labour shortages due to a rapidly shrinking labour force as more than 28% of the population is over 65 years old and less than 60% are between the ages of 15 and 64. However, the total unemployment rate of Japan remains at a low percentage of 2.6% as 80.1% of the population between 20 and 69 were employed, which leaves very little labour reserve (Liu-Farrer *et al.*, 2023). Moreover, Japan attempted to utilise labour from other countries to reduce labour shortages, however there were various elements that lessened the effectiveness of this approach:

- Japan has a selective immigration policy that enabled skilled professionals to join regular employment, however, manual or service workers were only allowed to work temporarily, about three to five years. Therefore, certain workers are unable to be employed by employers as regular employees due to restricted work permits or limited residence. This doesn't assist in alleviating labour shortages in industries like construction, health and agriculture.
- Japanese firms require specific skills such as proficiency in the Japanese language and cultural competency. Additionally, job applicants must complete a language evaluation test to qualify for interviews, where the criteria is strict enough that international students that graduate from Japanese universities have difficulty with it. Therefore, these criteria can prevent foreign workers from joining the job market in Japan.
- Japanese firms may have little to no experience with cultural diversity, which can influence them from recruiting foreign workers.
- Japanese firms provide on-the-job training; however, the skills developed with this training is not necessarily applicable to other companies and can dissuade foreign workers from working for these firms (Liu-Farrer *et al.*, 2023).

Therefore, robots are being utilised by various industries in Japan to reduce labour shortages, such as the hospitality industry. This industry utilises robots to improve labour productivity, simplify tasks for workers and assists ageing workers complete their tasks. Applications of robotics in these industries can include automated cooking as well as dish washing, picking and sorting, conveyor belts that automates the serving of food and a robot called "Pepper" that welcomes customer and receives data like the customer's preferred seating (Shibata Saori, 2019).

South Africa has the highest unemployment rates in the world where young people are more at risk to in comparison to adults. Additionally, the rates of unemployment between the years of 2008 to 2019 indicated that over half the youth population were experiencing it, which can act as an indication that this unemployment rate might continue to increase over time (Godfrey Manamela and Susan Ngomane, 2022). Moreover, robotics and AI-based systems can further widen the unemployment rate where workers can be replaced with automated machines and reshape the skills needed to perform existing tasks. Furthermore, any new job opportunities that robotics and AI-based systems can present might not contribute to solving the unemployment crisis due to the people lacking the necessary skills to operate and maintain this technology. South Africans will have to reskill and upskill through training programmes and educational initiatives, which is something that not everyone in South Africa has the opportunity to partake in due to being a part of marginalised communities (Mokofe, 2024).

Both Japan and South Africa are experiencing issues with regards to labour in their country that can impact their economy. Both countries are unable to utilise labour from other countries in order to assist with their labour issues, because Japan has strict migration policies and South African has an unemployment crisis. Additionally, both the labour issues that they are experiencing are likely to worsen in the future.

However, the labour issues that Japan and South Africa are experiencing are also distinct from each other. Japan doesn't have an unemployment issue like South Africa, they have a labour shortage issue where they aren't that many people left to fill up the remaining jobs, unlike South Africa where they struggle to get jobs for their population due to issues like lack of skills from unequal education across the entire country. Furthermore, robotics can be utilised to assist with the labour shortage in Japan by automating job positions that weren't going to be filled up with workers, while robotics in South Africa worsens the unemployment issue by automating jobs that people are trying to achieve.

However, there are other advantages associated with robotics in South Africa despite the unemployment issues such as:

- In the industry of construction, strategies that adopt robotics indicate an increase in productivity and efficiency for complex projects by enabling the robot to carry more workloads than the workers, which can minimise exhaustion, as well as perform more work than what a labourer can do normally (Boya *et al.*, 2022).
- In the automobile industry, robotics can be utilised to perform the repetitive tasks with higher precision and repeatability in comparison to human workers, which can reduce the amount of musculoskeletal disorders that a worker can experience from performing repeating actions for a prolonged period of time. Additionally, robots can reduce the risk of accidents occurring due to human error or self-afflicted harm by performing precarious tasks like lifting heavy loads, however it should be noted that close vicinity between human workers, and the robots can create unsafe situations such as collisions. Therefore, this technology should be constantly monitored and integrated in a manner where these situations can't or are unlikely to occur (Chigbu and Nekhwevha, 2022).
- Robotic-assisted surgery can be utilised in the healthcare industry for precision medicine in orthopaedics due to its accuracy and consistency. Furthermore, this technology can also be used to advance other areas in healthcare such as diagnostics, rehabilitation and patient monitoring. However, it's difficult to ensure that this technology is available to as many South African as possible without reinforcing geographic disparities. Therefore, this technology can be implemented through gradual decentralisation while utilising mobile units and simulation hubs in order to ensure that this technology can reach underserved areas (Hansen and Nyarko, 2025).

1.4) The sharing economy refers to the leveraging of idling goods, goods that aren't currently in use, by sharing it with others for monetary or other forms of compensation. The sharing economy differs from other forms of consumption like second-hand purchasing, trading and gifting, because the sharing economy facilitates the concept of "access over ownership" where the ownership of the goods aren't transferred and access to these goods are temporary. Additionally, aspects of the sharing economy, such as the provision of information, access to new markets and facilitating payment, are conducted through ICT like websites and applications. Moreover, the goods present in the sharing economy are tangible and can't be intangible due to tangible goods possessing clear idling capacity, clear mechanisms for access and ownership as well as being unable to be accessed simultaneously with other individuals or groups (Curtis and Lehner, 2019).

These idling goods typically refer to expensive assets where individuals find value in paying for temporary access to it. These assets and the use cases where individuals pay for temporary access to it consist of:

- The owner's only or spare vehicle that people want to utilise for short periods of time, such as a tourist renting a vehicle while in another country.
- The owner's home that are typically occupied by tourists when visiting other countries as a cheaper alternative to hotels.
- The owner's specialised tool set that an individual needs to solve an issue where purchasing the tool set is too expensive or that the issue is unlikely to occur again in the future.
- Professional camera equipment that will only be utilised for specific occasions where it's cheaper to rent than purchase it.

Furthermore, the platforms that are built around the concept of the sharing economy enables the owners of these underutilised goods to make an income due to the flexibility and low barriers to entry that these platforms provide. Moreover, these platforms also provide a cheaper alternative to advertising goods or services that an owner can provide in comparison to listing on websites or the owner creating and maintaining their own website. Therefore, the owners are able to save their income due to the low cost of advertising, which can also enable them to expand supplies during short peak demand periods (Wirtz *et al.*, 2019). In addition, these owners can further save their income, because the goods that they provide is already in their ownership where they don't have to pay for additional physical assets.

There are various applications that utilise the concept of the sharing economy include:

- Uber drivers utilise their vehicles to perform trips for customers that align with their preferred destination, such as routes to and from their place of employment. Therefore, Uber drivers can utilise their vehicles for additional tasks that can act as a primary or secondary source of income, where the earnings they receive can be higher than traditional taxi businesses (Wirtz *et al.*, 2019).
- Relayrides enables owners to advertise their vehicles to others, where they can set the price and the periods of time when their vehicle is available. Relayrides will then recommend the owner's vehicle to users that are located the closest to them to increase the likelihood that someone will rent the vehicle for as long as they need. In addition, 75% of the rent that the user pays for will be sent to the owner while the other 25% acts as a service charge (Petrini, de Freitas and da Silveira, 2026).
- Airbnb enables home owners to offer their homes to others while they are not currently occupying it during a period of time. The popularity of Airbnb has a negative impact on the

performance and occupancy of traditional accommodation sectors like hotels due to aspects like increased flexibility, value for money and a guest-host relationship (Hall *et al.*, 2022).

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